Multiphysics Analysis of a Convergent–Divergent Rocket Nozzle Using Analytical, RPA, CFD, and Thermo-Structural Methods

An independent propulsion analysis project integrating theoretical modeling, numerical and flow simulation, and structural integrity assessment of a high-temperature rocket nozzle.

Objective

To design and analyze a convergent–divergent rocket nozzle using analytical relations and RPA, validate flow behaviour through CFD, and evaluate thermal and structural response of Inconel 718 under realistic operating conditions. To test a real rocket propulsion behaviour in a nozzle.

Literature Reference

Relevant literature on coupled flow–thermal analysis of rocket nozzles was reviewed to understand established multi-physics methodologies. In particular, the work by Sheikh & Kumar (2024) on coupled flow–thermal nozzle analysis informed the coupling approach, while all geometry, operating conditions, and results in the present study were independently defined and obtained.

RPA-Based Design & Results

Classical compressible flow relations were used to determine throat conditions, expansion ratio, and ideal performance parameters. Rocket Propulsion Analysis (RPA) was employed to estimate chamber pressure, mixture ratio, combustion properties, and ideal thrust parameters, serving as a baseline for validation.

Icame up with rocket nozzle design, initial parameters by research and analytical calculation which I gave as input to RPA and below are the parameters which I gave RPA

Chamber Pressure	30 bar
Propellant Used	RP-1 (fuel) + LOX (oxidizer)

Mixer Ratio	2.784 (fuel rich) Conical Nozzle with 15
Nozzle Shape	degree divergent half angle
Area Ratio for Convergent Nozzle	6.25
Area Ratio for Divergent Nozzle	12.25
Diameter of throat	10 cm
Diameter of chamber	25 cm
Diameter at nozzle exit	35 cm
Length (from chamber start to nozzle exit)	75 cm
Liquid Oxygen (LOX) at storage tank	Temperature = 86 k Pressure = 4 bar Mass fraction = 0.74
RP-1 Fuel at Storage Tank	Temperature = 293 K Pressure = 3 bar Mass Fraction = 0.26

Propellant Specification

Component	Temperature (K)	Mass fraction	Mole fraction
RP-1	293.00	0.26	0.45
O2(L)	86.00	0.74	0.55
	Total	1.00	1.00

Exploded propellant formula: $O_{1.097}\ C_{0.451}\ H_{0.880}$

O/F = 2.784

O/F^0 = 3.406 (stoichiometric)

α_{ox} = 0.817 (oxidizer excess coefficient)

Temperature (K): Temperature at which propellants are stored in tanks

Rocket Propulsion Analysis (RPA) Results

So, this is the result what I got from RPA, there are two performances I got for rocket nozzle, theoretical (ideal) performance and estimated delivered performance. For this project, I had taken estimated delivered performance as it also includes losses and tells nearly accurate parameters what we need for our analysis to understand more clearly about rocket nozzle flow, structural and thermal behaviour.

Through results, I formulate my further journey including selecting which type of flow, what kind of material should I consider, and what are the assumptions I should take for this project.

Table 1. Thermodynamic properties

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit	Unit
Pressure	3.0000	2.9692	1.7296	0.0364	MPa
Temperature	3567.0178	3564.6476	3403.4467	2461.0488	K
Enthalpy	-772.1112	-778.5685	-1430.8459	-5136.3004	kJ/kg
Entropy	11.3792	11.3810	11.3810	11.3810	kJ/(kg·K)
Internal energy	-2017.4873	-2023.0257	-2601.5706	-5912.6561	kJ/kg
Specific heat (p=const)	7.6970	7.7024	7.6671	4.0383	kJ/(kg·K)
Specific heat (V=const)	6.4770	6.4820	6.5040	3.5476	kJ/(kg·K)
Gamma	1.1884	1.1883	1.1788	1.1383	
Isentropic exponent	1.1297	1.1296	1.1254	1.1273	
Gas constant	0.3491	0.3491	0.3440	0.3155	kJ/(kg·K)
Molecular weight (M)	23.8144	23.8161	24.1712	26.3569	
Molecular weight (MW)	0.02381	0.02382	0.02417	0.02636	
Density	2.4089	2.3859	1.4774	0.0468	kg/m³
Sonic velocity	1186.1068	1185.6329	1147.8168	935.5280	m/s
Velocity	0.0000	113.6424	1147.8168	2954.3829	m/s
Mach number	0.0000	0.0958	1.0000	3.1580	
Area ratio	6.2500	6.2500	1.0000	12.2500	
Mass flux	271.1425	271.1425	1695.7579	138.3819	kg/(m²·s)
Mass flux (relative)	0.904e-04	0.913e-04			kg/(N·s)

Table 3. Theoretical (ideal) performance

Parameter	Sea level	Optimum expansion	Vacuum	Unit
Characteristic velocity		1760.06		m/s
Effective exhaust velocity	2484.95	2954.38	3217.16	m/s
Specific impulse (by mass)	2484.95	2954.38	3217.16	N·s/kg
Specific impulse (by weight)	253.39	301.26	328.06	s
Thrust coefficient	1.4119	1.6786	1.8279	

Table 4. Estimated delivered performance

Parameter	Sea level	Optimum expansion	Vacuum	Unit
Characteristic velocity		1716.47		m/s
Effective exhaust velocity	2321.69	2791.13	3053.91	m/s
Specific impulse (by mass)	2321.69	2791.13	3053.91	N·s/kg
Specific impulse (by weight)	236.75	284.62	311.41	s
Thrust coefficient	1.3526	1.6261	1.7792	

Ambient condition for optimum expansion: H=7.86 km, p=0.359 atm

Table 5. Altitude performance

Altitude, km	Pressure, atm	Effective exhaust velocity, m/s	Specific impulse (by weight), s	Thrust coefficient
0.000	1.00000	2484.951	253.394	1.4119

Calculation:

1) Thrust (F) = Cf * Pc * At = 1.3526 * 30e5 * 0.07854 = 31869.89 N

   where, Pc = Chamber pressure = 30 bar

At = throat cross-sectional area = 0.00785 m^2
Cf = coeƯicient of thrust = 1.3526

2) Mass Flow Rate (m): F=m*Vj + (Pe – Pa) * Ae

m=(F + (Pa - Pe) * Ae)/Vj = 16.41 kg/s

where,Vj=EƯectiveExhaust Velocity = 2321.69 m/s

Pa=AmbientPressure = 1.01325 bar

Pe=Pressureatnozzle exit = 0.364 bar

Ae=Cross-sectional area at nozzle exit = 0.096 m^2

Summary of result, I got from RPA

EƯective Exhaust Velocity	2321.69 m/s
Specific Impulse	236.75
Thrust CoeƯicient	1.3526
Thrust	31869.89 N
Mass Fow Rate	16.41 kg/s

CFD Flow Analysis

Axisymmetric CFD simulations were performed to resolve compressible flow through the nozzle. Mach number, pressure, density, and velocity contours were analyzed to identify choked flow at the throat, expansion behaviour, and shock structures (Mach diamonds) in the divergent section. Numerical thrust values were extracted and compared with analytical and RPA predictions.

Geometry:

Setup:

Solver Type: Density-based, transient 2D axisymmetric with absolute velocity
Fluid Material Used: RP-1 and LOX mixture

Properties: Density = Idea Gas
Specific Heat at constant pressure (Cp) = piecewise-linear
Thermal Conductivity = constant
Viscosity = Sutherland
Molecular weight = 23.8144 kg/mol

Boundary Conditions:

Blue arrows show inlet (fluid flows from it)

Red arrows show outlet (fluid flows to it)

Inlet Absolute Pressure = 30 bar

Inlet Absolute Temperature = 3567.0178 K

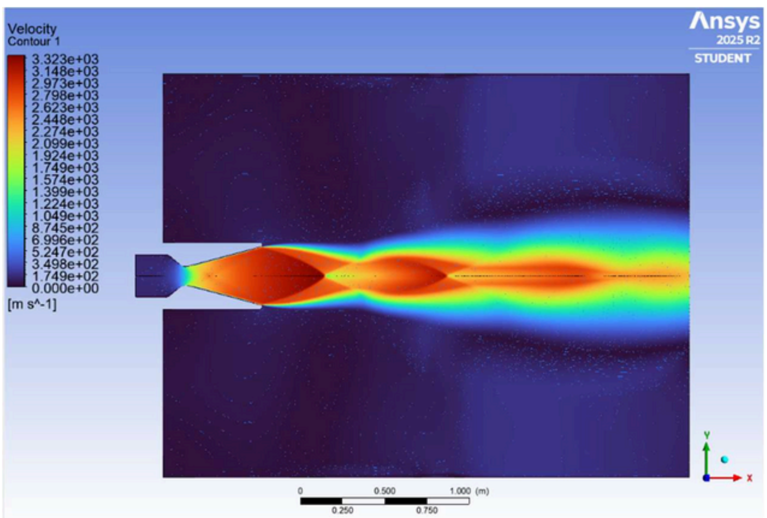
Outlet Absolute Pressure = 1.01325 bar

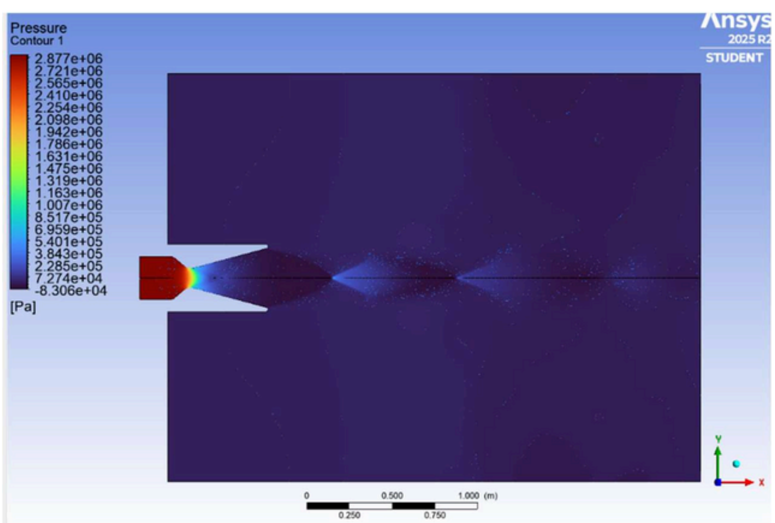
Outlet Absolute Temperature = 300 K

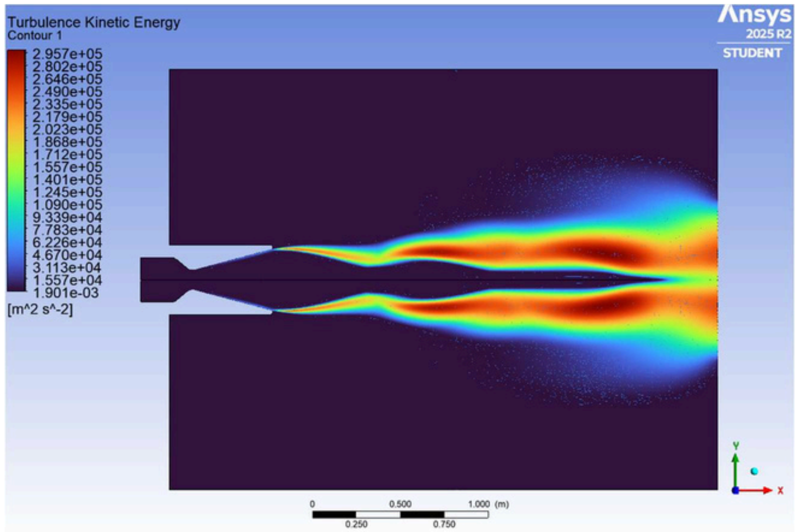
CFD Results:

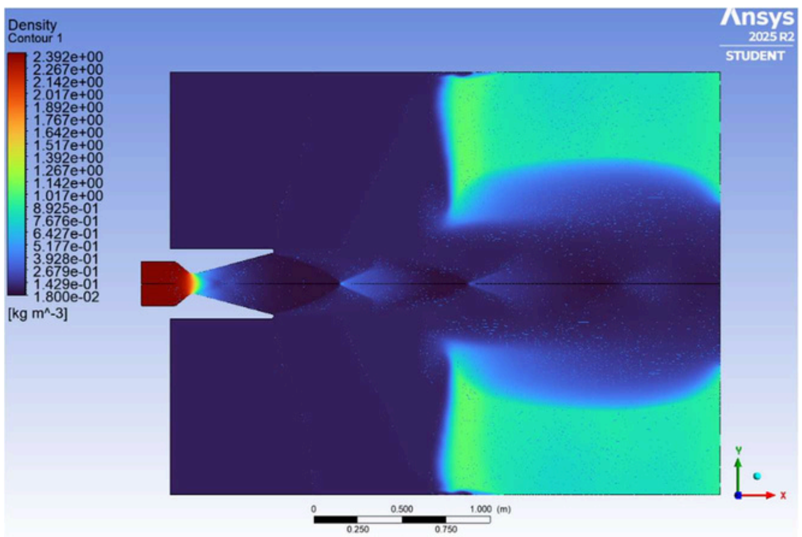
Expected: Mach diamond formation, choked flow at nozzle throat, thrust, specific impulse, eUective exhaust velocity from CFD do not vary too much with actual results that we got from RPA and analytical calculation.

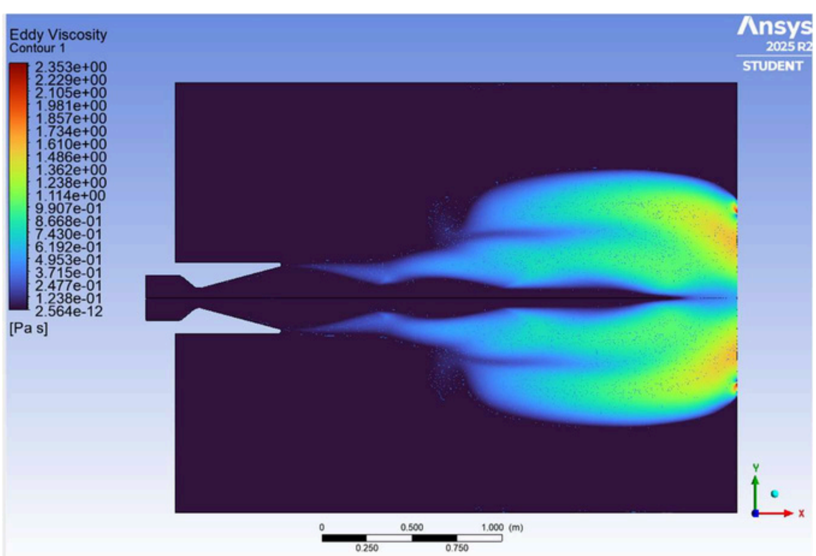
Here are some contours of important parameters:






Turbulence Kinetic Energy
Contour 1
2.957e+05
2.802e+05
2.646e+05
2.490e+05
2.335e+05
2.179e+05
2.023e+05
1.868e+05
1.712e+05
1.557e+05
1.401e+05
1.245e+05
1.090e+05
9.339e+04
7.783e+04
6.226e+04
4.670e+04
3.113e+04
1.557e+04
1.901e-03
[m^2 s^-2]
Ansys
2025 R2
STUDENT
0
0.500
1.000 (m)
0.250
0.750


Density
Contour 1
2.392e+00
2.267e+00
2.142e+00
2.017e+00
1.892e+00
1.767e+00
1.642e+00
1.517e+00
1.392e+00
1.267e+00
1.142e+00
1.017e+00
8.925e-01
7.676e-01
6.427e-01
5.177e-01
3.928e-01
2.679e-01
1.429e-01
1.800e-02
[kg m^-3]
Ansys
2025 R2
STUDENT
0
0.500
1.000 (m)
0.250
0.750


Eddy Viscosity
Contour 1
2.353e+00
2.229e+00
2.105e+00
1.981e+00
1.857e+00
1.734e+00
1.610e+00
1.486e+00
1.362e+00
1.238e+00
1.114e+00
9.907e-01
8.668e-01
7.430e-01
6.192e-01
4.953e-01
3.715e-01
2.477e-01
1.238e-01
2.564e-12
[Pa s]
Ansys
2025 R2
STUDENT
0
0.500
1.000 (m)
0.250
0.750

Results got from CFD: These values are approximated (not exact)

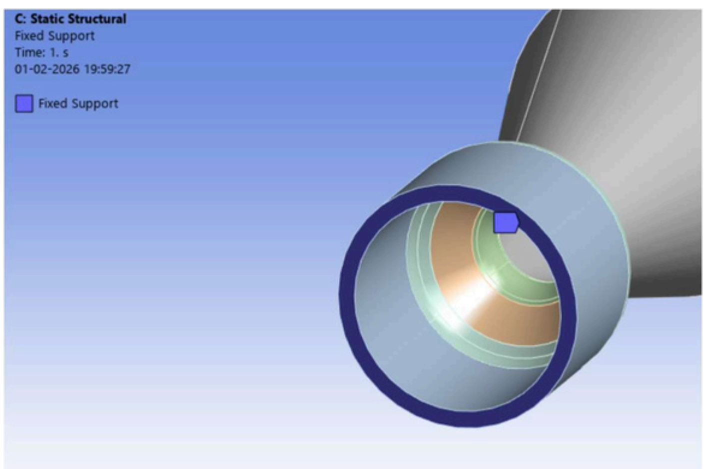
EUective Exhaust Velocity	2660 m/s
Mass flow rate	15 kg/s
Thrust	33653.48 N
Specific Impulse	271.15 s
CoeUicient of Thrust	1.428

Structural Analysis:

Structural analysis was performed using Inconel 718 as the nozzle material. Von-Mises stress distribution, and total deformation got from which tested material strength.

S etu p:

Fixed Support: at nozzle inlet surface



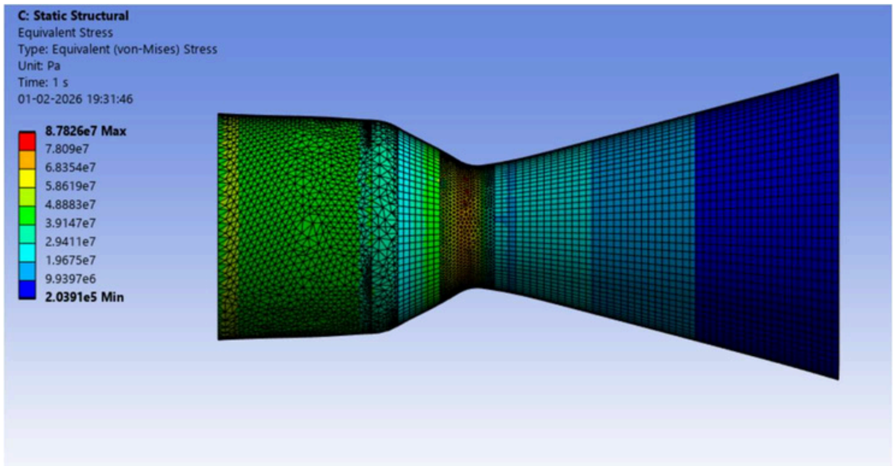
Pressure imposed on inner surfaces of converging nozzle and chamber section = 30 bar

Pressure imposed on inner surfaces of throat = 17.296 bar

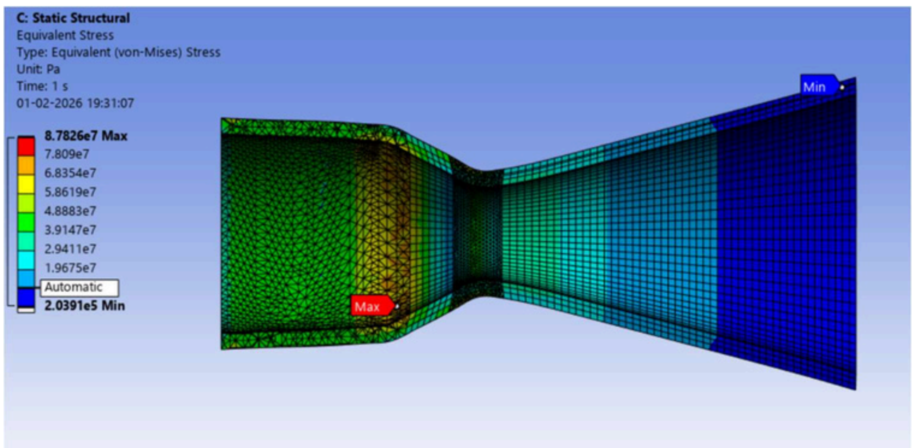
Pressure imposed on inner surfaces of diverging nozzle = 8.83 bar

Results:

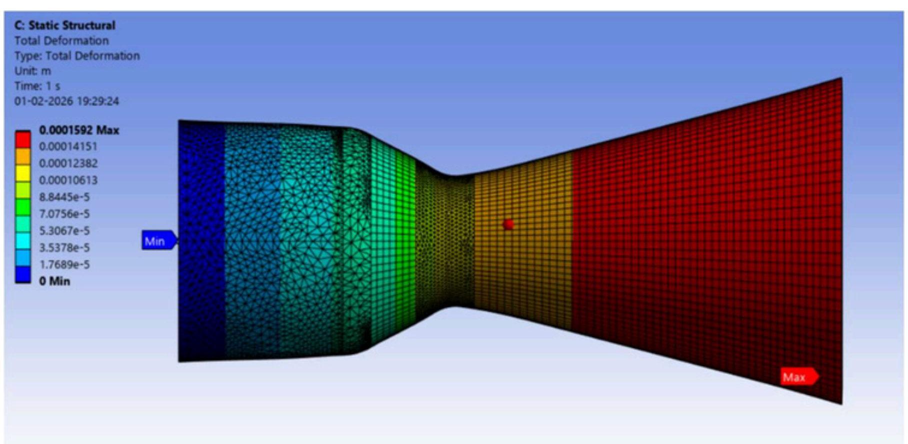
These are contours of von-miss stress (equivalent stress) and total deformation.



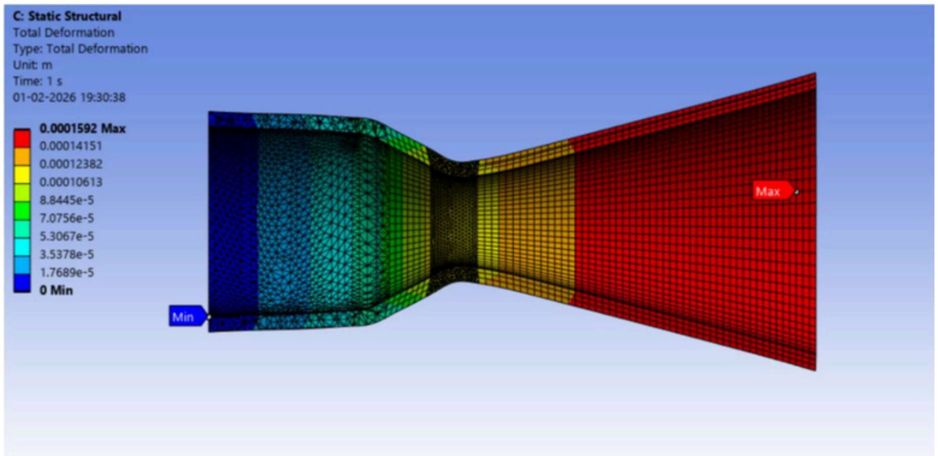
Outer surface Contour



Inner Surface Contour



Outer Surface Contour



Inner Surface Contour

Calculation:

Initial diameter of nozzle exit = 35 cm

Deformation at nozzle exit = 0.0001592

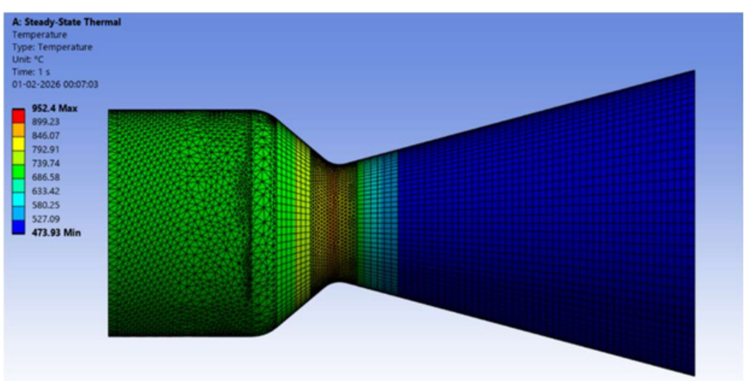
Final diameter of nozzle exit = 35 * 0.0001592 + 35 = 35.0056 cm

Since deformation is not too much so area ratios will not change much and all parameters are nearly same as initial values.

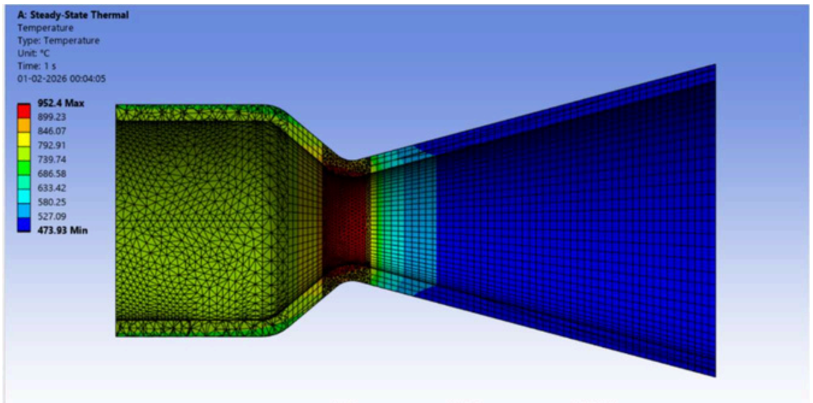
Thermal Analysis:

A steady-state thermal analysis was performed to estimate the temperature distribution and heat transfer trends in the convergent–divergent rocket nozzle under high-temperature gas flow conditions. The study identifies regions of maximum thermal loading, particularly near the throat, and provides insight into the thermal environment experienced by the nozzle material.

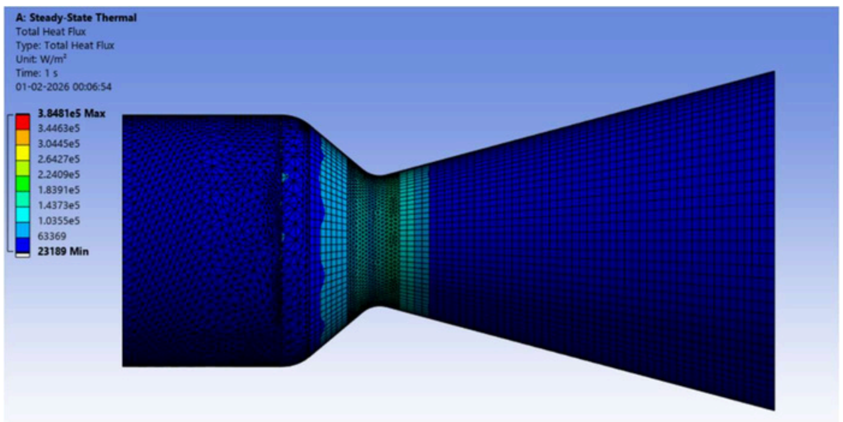
These are some contours of heat flux and temperature



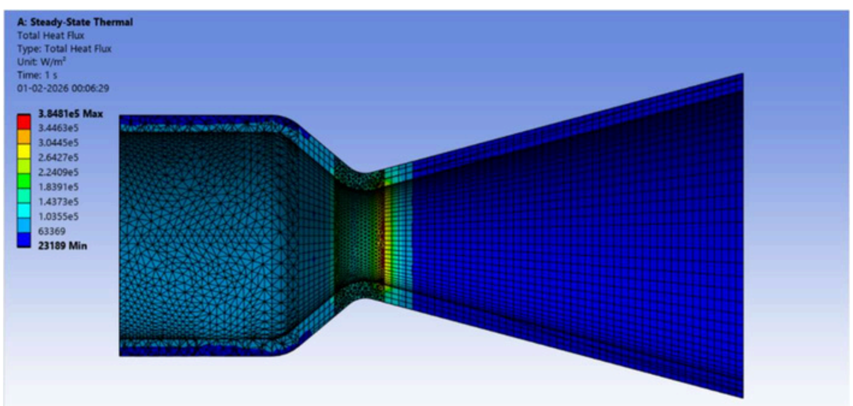
Outer Surface Contour



Inner Surface Contour



Outer Surface Contour



Inner Surface Contour

Since Melting Temperature of Inconel 718 is 1260 deg Celsius but here max temperature reaching is 960 deg ce

Since Melting Temperature of Inconel 718 is 1260 deg Celsius but here max temperature reaching is 960 deg Celsius (approx.) which means this material can be used for rocket nozzle.

Validation:

Results obtained from CFD were cross-validated against analytical calculations and RPA outputs. Flow choking, thrust levels, and pressure trends showed strong agreement across methods, providing confidence in the numerical and structural results.

Conclusion:

This project demonstrates a complete rocket nozzle analysis workflow, integrating analytical modeling, RPA, CFD, structural and thermal simulations. The study highlights the importance of multi-method validation and provides insight into flow behaviour, heat transfer, and structural response of high-temperature rocket nozzles.